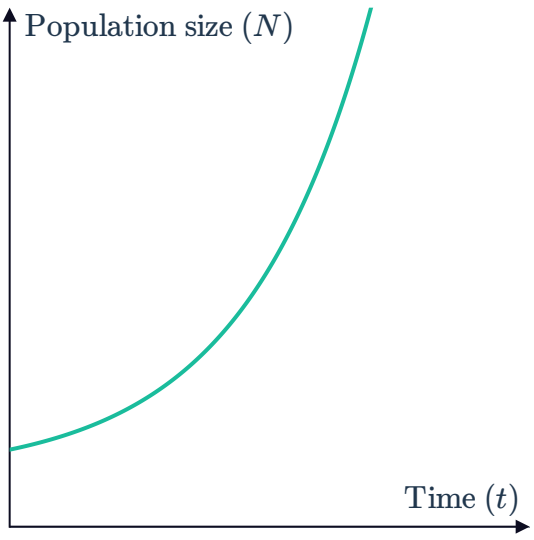
Linear and exponential functions

1 Identify the type of function shown in the following graph:



2 Two different sequences are generated by the functions $f(n) = 5n + 1$ and $g(n) = 3(2)^n$.

a Find the first 5 terms in the sequence generated by each function:

n	1	2	3	4	5
$f(n)$					
$g(n)$					

b Describe how the terms of each sequence increase.

c Is John correct to conclude that the terms of $g(n)$ will always be greater than the terms of $f(n)$ for $n > 1$? Explain your answer.

3 Two functions, $q(x) = (1.09)^x$ and $r(x) = 39x + 145$, both have domains of $x > 0$.
For what type of values of x is $q(x) > r(x)$?

4 Consider the following table of values for $f(x) = 5^x$ and $g(x) = 2190x$ for $x \geq 1$:
Neville concludes that $g(x)$ is always greater than 5^x when $x \geq 1$.
Determine whether Neville is correct, and explain your answer.

x	$f(x)$	$g(x)$
1	5	2190
2	25	4380
3	125	6570
4	625	8760
5	3125	$10\,950$

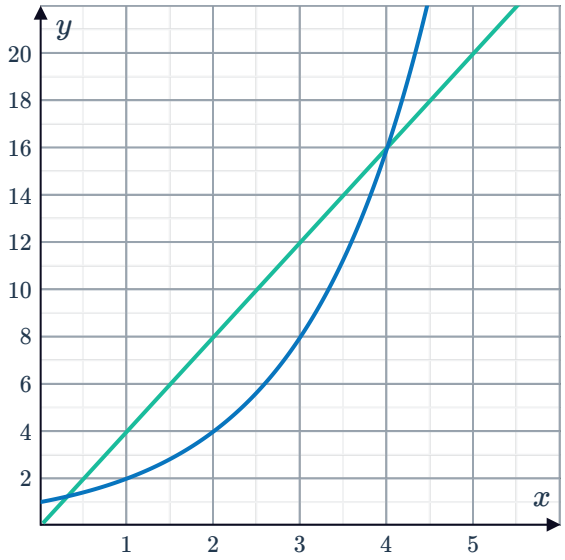
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5 Consider the table of values for the functions $f(x) = (1.05)^x$ and $g(x) = 5x$ for $x \geq 1$.

x	1	2	3	4	5
$f(x)$	1.05	1.10	1.16	1.22	1.28
$g(x)$	5	10	15	20	25

Ivan concludes that $g(x)$ is always greater than $(1.05)^x$ for all values of $x \geq 1$. Determine whether Ivan is correct, and explain your answer.

6 A linear function and exponential function have been graphed on the same coordinate plane:



a For each 1 unit increase in x, by how much does the linear function increase?

b For each 1 unit increase in x, by what multiplicative factor does the exponential function increase?

c As x approaches infinity, which function increases more rapidly?

7 Consider the functions $f(x) = 3x$ and $g(x) = 3^x$.

a Complete the table of values for each function:

x	0	1	2	3	4
$f(x)$					
$g(x)$					

b Sketch the graph the functions on the same coordinate plane.

c

i For every 1 unit increase in x, the y value of $f(x)$ increases by how many units?

ii For every 1 unit increase in x, the y value of $g(x)$ increases by a factor of what?

d For $x \geq 1$, which function increases at a faster rate?

e Find the following function values:

i $f(8)$ **ii** $g(8)$

f Describe the behavior of each function as x increases

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8 Consider the linear function $f(x) = 5x + 2$ and the exponential function $g(x) = 5(3)^x$.

a Complete the table of values:

x	4	5	6
$f(x)$			
$g(x)$			

b Evaluate:

i $f(5) - f(4)$

ii $\dfrac{g(5)}{g(4)}$

iii $f(k+1) - f(k)$

iv $\dfrac{g(k+1)}{g(k)}$

c For each 1 unit increase in x, describe how the linear and exponential functions increase.

9 Consider the exponential function P given by the equation $y = 5^{-x} + 3$ and the linear function Q given by the equation $y = -4x + 4$.

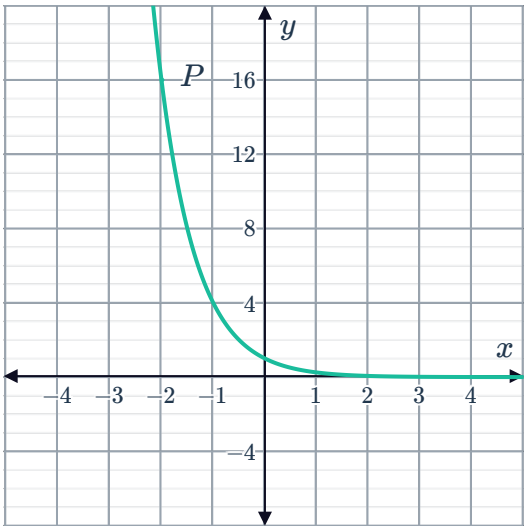
a Sketch the graph of P.

b On the same coordinate plane, sketch the graph of Q.

c How many times do P and Q intersect?

d As x approaches ∞, which function is greater?

10 The graph of the exponential function P, given by $y = 4^{-x}$ is shown on the right:



a Sketch the graph of Q given by the equation $y = 8x + 12$ on the same coordinate plane.

b How many times do P and Q intersect?

c Which function has the greater function value at the y-intercept?

d Which function has the greater function value at $x = 1$?

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11 A table of values for the function P and a table of approximate values for the function Q is provided below:

- Function P

x	-4	-3	-2	-1	0	1	2
y	-3	0	3	6	9	12	15

- Function Q

x	-5	-2	-1	0	1	5	10
y	$-32\,765$	-61	-5	2	$2\frac{7}{8}$	$2\frac{32\,767}{32\,768}$	$2\frac{1\,073\,741\,823}{1\,073\,741\,824}$

a Which of the functions are linear and non-linear?

b Sketch the graph of function P.

c Given that Q is an exponential function, sketch the graph of the function Q on the same coordinate plane.

d Using part (c), how many solutions are there to the equation $P - Q = 0$?

12 Consider the following pair of functions:

- Function P
  $y = 6^{-x} + 5$

- Function Q

x	-4	-3	-2	-1	0
y	0	$-\frac{11}{4}$	$-\frac{11}{2}$	$-\frac{33}{4}$	-11

a Determine what type of functions P and Q are.

b Sketch the graph the functions on the same coordinate plane.

c Identify which function has the greater function value at the y-intercept.

d Identify how many times P and Q intersect.

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Applications

13 During a sudden outbreak, scientists must decide between two anti-bacterial treatments that are currently being trialed to try to control the outbreak. In the laboratory, they apply Treatment A and Treatment B to two samples of the bacteria, each containing 300 microbes. They keep track of the number of microbes in each sample. The table shows the results:

Number of hours (t)	0	2	4	6
Number of microbes using Treatment A	300	310	320	330
Number of microbes using Treatment B	300	900	2700	8100

a Which treatment causes the number of microbes to increase linearly?

b By what amount is the number of microbes increasing each hour using Treatment A?

c Which treatment will better control the number of microbes?

d Scientists approximate that they'll have a more effective treatment in 10 hours. By that time, what will be the difference between the number of microbes in the two samples?

14 Sophia is considering two different career options:

- She can accept a job with a company that has offered her a salary of $\$125\,000$ a year with a projected salary increase of $\$3000$ per year.
- She can start her own business where she estimates her initial annual salary to be $\$20\,000$ per year with a projected annual salary increase of 40% each year.

a Form an equation for $f\,(n)$, her salary in n years time if she accepts the job with a company.

b Form an equation for $g\,(n)$, her salary in n years time if she starts her own business.

c Complete the table of values:

n	2	3	4
Increase in salary compared to previous year when working for a company			
Increase in salary compared to previous year when working for her own business			

d After 3 years, which option will result in a greater annual salary for Sophia?

e After 6 years, which option will result in a greater annual salary for Sophia?

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15 When comparing two phone companies, Derek notices that both offer a $\$200$ monthly contract, but they have different policies on how they charge fees if the bill is overdue:

- Company A: Start adding $\$6$ to the bill for every day the bill is overdue.
- Company B: Increase the bill by 4% per day that the bill is overdue.

a Complete the table of values for the total amount of the bill if it is x days overdue under each company's policy.

x	0	3	6	9
Company A's bill	200			
Company B's bill	200			

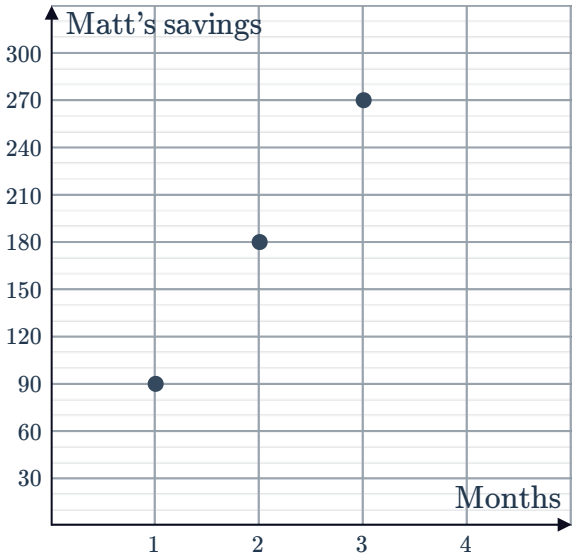
b If an unpaid bill reaches $\$3000$, it is referred to a debt collector. Under which company's policy would an unpaid bill first reach $\$3000$?

16 Matt and Sophia are saving money using different strategies. The amount each has saved after each month is given by the table of values and the plotted points.

- Sophia:

Number of months	1	2	3
Sophia's savings	4	16	64

- Matt:



If each person continues their pattern of saving, determine who will be the first to exceed savings of $\$600$.

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17 During a sudden viral outbreak, scientists must decide between two antivirals to try and control the situation. In a laboratory, they apply Adravil and Felicium to two samples of the virus, each containing 200 microbes.
They keep track of the number of microbes in each sample, and notice that the number of microbes using Adravil is increasing by a constant amount of 12 each hour. The table shows the results for Felicium.

Number of hours (t)	0	3	6	9
Number of microbes using Felicium	200	600	1800	5400

a Determine which antiviral will better control the number of microbes. Explain your choice.

b The new antiviral, Tretonin, shows the following preliminary results:

Number of hours (t)	0	3	6	9
Number of microbes using Tretonin	200	202	208	218

If the trends from the first 9 hours continue in the future, determine which treatment will be better of the short-term, and which will be better over the long-term. Explain your answer.

18 A bakery purchased $\$800$ in new appliances to open up a new store. Every year thereafter, the appliances depreciated by 50% of their previous year's value. Let V be the sale value of the appliances t years after its purchase.

a Plot the points on a coordinate plane that relate to the initial value of the appliances and the sale value after 4 years.

b Is the sale value of the appliances decreasing exponentially or linearly?

c Form an equation for the sale value of the appliances, V, after t years.

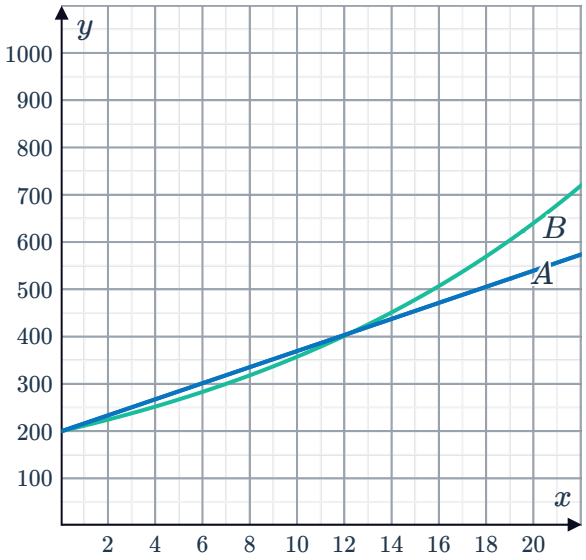
d Sketch the graph of the function.

e By how much will the sale value after 8 years differ from the sale value after 4 years?

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19 When comparing two phone companies, Mohamad notices that both offer a $\$200$ monthly contract, but they have different policies on how they charge fees if the bill is overdue:

- Massive Dynamic's policy is to add $\$17$ to the bill for every day the bill is overdue.
- Stark Industry's policy is to increase the bill by 6% for every day the bill is overdue.



a Complete the table of values for the total amount of the bill if it is x days overdue under each company's policy:

x	0	4	8	12
Massive Dynamic's bill				
Stark Industry's bill				

b The two policies are graphed above. Which graph represents which policy?

c If an unpaid bill reaches $\$3000$, it is referred to a debt collector. Under which company's policy would an unpaid bill first reach $\$3000$?

d Mohamad signs up for Stark Industry, and is sent a bill for $\$200$. He leaves the country a few days later, and when he comes back home the bill is 80 days overdue. How much does he owe? Give your answer to the nearest dollar.

e What could Mohamad do to prevent such expensive long overdue bills in the future?

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20 Sharon is considering two different career options:

- She could accept a job with Kord Enterprise that has offered her an income of $\$125\,000$ for the first year, and a projected income increase of $\$49\,000$ each year after that.
- She can start her own business, where she estimates her initial annual income to be $\$40000$ for the first year, increasing by 60% each year after that.

a Complete the following table of values:

n	1	2	3	4
Income in year n with Kord Enterprise	125 000			
Income in year n with her own business	40 000			

b The two incomes are graphed on the right:
Which graph represents which income?



c In the 5th year, which option will result in a greater annual income for Sharon?

d In the 10th year, which option will result in a greater annual income for Sharon?

e Is a salary rise of 60% realistic in the long run?

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